

Reliable Classification

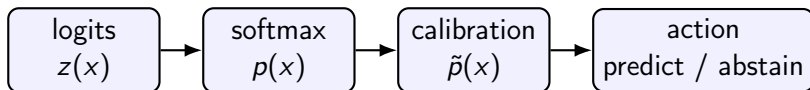
Calibration, proper scoring, post-hoc repair, uncertainty, and abstention

Graduate Deep Learning

Lecture structure

- 1 Reliability pipeline and split discipline
- 2 What calibration means
- 3 Scoring and auditing probabilities
- 4 Post-hoc calibration
- 5 Uncertainty beyond a single softmax
- 6 Selective prediction and abstention
- 7 Beyond IID reliability

Reliable classification is a pipeline



- A deployed classifier is not only a label generator.
- It produces **probabilities** and then uses them for a **decision**.
- Reliability studies the whole chain: honest numbers, useful uncertainty, and sensible action.
- A confidence of “0.98” is meaningful only if that number is numerically trustworthy.

Bottom line: accuracy is only one part of reliability.

Running examples and notation

Example A: binary weather table

Day	p	y
1	0.90	1
2	0.70	0
3	0.40	1
4	0.20	0

Used later for **Brier score**, **log loss / NLL**, and the appendix weather table.

Example C: same correctness, different confidence

i	1	2	3	4	5	6	7	8
r_i	1	1	0	1	1	0	1	0
$c^{(A)}$	0.95	0.90	0.80	0.75	0.60	0.55	0.40	0.35
$c^{(B)}$	0.70	0.70	0.70	0.70	0.55	0.55	0.55	0.55

Used later to show why **ECE alone can mislead** even when accuracy matches.

Example B and notation

Example B is the main 8-row multiclass table; it appears on the **next slide** and returns for the **softmax example**, **worked ECE**, **risk-coverage**, and the appendix multiclass tables.

$z(x) \in \mathbb{R}^K$: logits; $p(x) = \text{softmax}(z(x))$: predicted probabilities; $\hat{y}(x) = \arg \max_k p_k(x)$: predicted class; $c(x) = \max_k p_k(x)$: top-label confidence; $\tilde{p}(x)$: calibrated probability vector.

Why it matters: the same objects reappear in scoring, repair, and abstention.

Tiny multiclass running example (Example B)

#	Pred.	Conf.	True	Correct?
1	A	0.894	A	1
2	A	0.696	A	1
3	B	0.777	B	1
4	A	0.595	B	0
5	C	0.812	C	1
6	A	0.746	B	0
7	A	0.426	C	0
8	B	0.500	C	0

- Accuracy = $4/8 = 0.50$.
- Mean top confidence ≈ 0.681 .
- Since $0.681 > 0.50$, the model looks overconfident on this toy set.

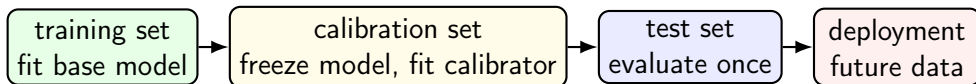
One explicit softmax calculation

For $z = (3.5, 1.0, 0.2)$,

$$p_A = \frac{e^{3.5}}{e^{3.5} + e^{1.0} + e^{0.2}} \approx 0.894.$$

- **Symbols:** z are raw class scores; softmax converts them to probabilities.
- **Interpretation:** the largest logit becomes the largest probability.
- **Why it matters:** calibration and abstention operate on these probabilities, not directly on logits.

Clean split discipline



- Train the classifier first.
- Freeze it before post-hoc calibration.
- Fit the calibrator only on a separate holdout set.
- Evaluate the final calibrated system once on the test set.

Common mistake

Do *not* fit the calibrator on the test set. That leaks information and makes reliability look better than it is.

Calibration in one sentence

Binary calibration

$$\mathbb{P}(Y = 1 \mid \hat{p}(X) = u) = u.$$

- $Y = 1$: the target event/class happens, i.e. the “positive” case.
- X : a random input; one realized sample is written x .
- $\hat{p}(X)$: the model’s predicted chance that $Y = 1$; for one concrete sample x , we write $\hat{p}(x)$.
- $u \in [0, 1]$: a forecast value such as 0.2, 0.5, or 0.8.

Plain-English meaning

- Across many samples x for which the model outputs about 0.8, about 80% should truly be positive.
- Calibration asks whether probability numbers match **observed frequencies** in the long run.
- It is not about one prediction being right or wrong.

Takeaway: calibration checks whether predicted probabilities are numerically honest over many inputs.

Calibration in one sentence: concrete example

Suppose 100 samples have $\hat{p}(x)$ near 0.8

X is a random input and x is one realized sample;
positive means $Y = 1$, the event/class of interest.

Scenario	Positives	Interpretation
A	80/100	calibrated at 0.8
B	60/100	over-confident
C	95/100	under-confident

Plug in $u = 0.8$

$$\mathbb{P}(Y = 1 \mid \hat{p}(X) = 0.8) = 0.8$$

So if one sample x gets $\hat{p}(x) = 0.8$, the model assigns an 80% chance to $Y = 1$. Across many such samples x , about 80% should truly have $Y = 1$.

Why this is a population statement

One example is either positive or negative. Calibration asks what fraction are positive *across many examples that received the same forecast*.

- If only 60 out of 100 are positive, the model is too sure.
- If 95 out of 100 are positive, the model is too cautious.
- The “100” means the samples whose predicted probabilities $\hat{p}(x)$ fall in a bin near 0.8.

Three calibration notions you must separate

Binary

$$\mathbb{P}(Y = 1 \mid \hat{p}(X) = u) = u.$$

Used when there are two classes and the forecast is one scalar probability.

Top-label / confidence

$$\mathbb{P}(Y = \hat{y}(X) \mid c(X) = u) = u.$$

Here $\hat{y}(X) = \arg \max_k p_k(X)$
and $c(X) = \max_k p_k(X)$.

Classwise / vector-aware

$$\mathbb{P}(Y = k \mid p_k(X) = u) = u.$$

Operational version of stronger multiclass calibration: every class probability should be honest, not only the largest one.

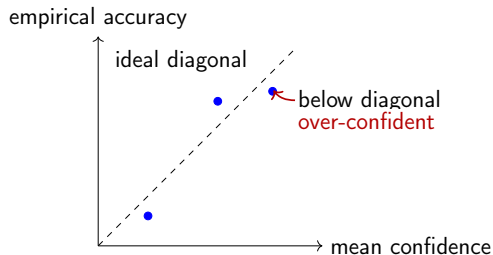
Bottom line: say explicitly which notion a metric or method targets.

Accuracy, calibration, and sharpness are different

Concept	Question answered	Key caution
Accuracy	Is the top class correct?	A model can be accurate and still attach bad probability values.
Calibration	Are the probabilities honest?	A model can be calibrated but uninformative.
Sharpness / resolution	Does the model separate easy and hard cases by confidence?	A model can look calibrated by collapsing forecasts toward the mean.

- Constant 0.5 forecasts on a balanced binary task can be calibrated but useless.
- Example C will show the opposite failure: two models have the same accuracy, but one uses confidence much more effectively.

Reliability diagrams



- Bin predictions by confidence.
- For each bin, plot **mean confidence** against **empirical accuracy**.
- Points on the diagonal are perfectly calibrated.
- Points *below* the diagonal: confidence exceeds accuracy (overconfidence).
- Points *above* the diagonal: confidence is too low (underconfidence).

Why it matters: a diagram shows *where* miscalibration occurs; a single scalar often hides that structure.

Proper scoring rules: the theoretical root

Definition

A scoring rule $S(p, y)$ is **proper** if the true label distribution minimizes expected score, and **strictly proper** if that minimizer is unique.

$$\mathcal{R}_S(p; q) = \sum_{k=1}^K q_k S(p, k), \quad \mathcal{R}_S(q; q) \leq \mathcal{R}_S(p; q).$$

- **Symbols:** p is the announced distribution; q is the true conditional label distribution; k indexes the class; K is the number of classes.
 - **Interpretation:** the best strategy is to forecast the truth.
 - **Why it matters:** proper scores define what honest probability estimation means.
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- The most important examples here are **Brier score** and **log loss**.
 - ECE is not a proper score; it is a descriptive calibration statistic.

Brier score

Binary Brier score

$$S_{\text{Br}}(a, y) = (a - y)^2, \quad L_{\text{Br}}(a) = \eta(1 - a)^2 + (1 - \eta)a^2.$$

- **Symbols:** $a \in [0, 1]$ is the forecast, $y \in \{0, 1\}$ is the outcome, $\eta = P(Y = 1 \mid X = x)$.
- **Interpretation:** squared distance between forecast and outcome.
- **Why it matters:** $L_{\text{Br}}(a)$ is the previous slide's $\mathcal{R}_S(p; q)$ in binary form, with $p = (a, 1 - a)$ and $q = (\eta, 1 - \eta)$.

$$\begin{aligned} \frac{dL_{\text{Br}}}{da} &= 2(a - \eta) = 0 \quad \Rightarrow \quad a = \eta, \\ \text{BS} &= \frac{(0.90 - 1)^2 + (0.70 - 0)^2 + (0.40 - 1)^2 + (0.20 - 0)^2}{4} \\ &= \frac{0.01 + 0.49 + 0.36 + 0.04}{4} = 0.225. \end{aligned}$$

Log loss / negative log-likelihood

Binary log loss

$$S_{\log}(a, y) = -y \log a - (1 - y) \log(1 - a),$$

$$L_{\log}(a) = -\eta \log a - (1 - \eta) \log(1 - a).$$

- **Symbols:** same a, y, η as before.
- **Interpretation:** being confidently wrong is punished sharply.
- **Why it matters:** it is the training loss behind cross-entropy classifiers.

$$\frac{dL_{\log}}{da} = -\frac{\eta}{a} + \frac{1 - \eta}{1 - a} = 0 \quad \Rightarrow \quad a = \eta,$$

$$\begin{aligned} \text{NLL} &\approx \frac{-\log 0.90 - \log 0.30 - \log 0.40 - \log 0.80}{4} \\ &\approx \frac{0.105 + 1.204 + 0.916 + 0.223}{4} \approx 0.612. \end{aligned}$$

Murphy decomposition: the big picture

Binary Brier decomposition

$$\text{BS} = \underbrace{\bar{y}(1 - \bar{y})}_{\text{uncertainty}} - \underbrace{\text{Var}(Q)}_{\text{resolution}} + \underbrace{\mathbb{E}[(Q - C)^2]}_{\text{reliability}}.$$

- $C \in [0, 1]$ is the forecast, $Q = \mathbb{E}[Y \mid C]$, and $\bar{y} = \mathbb{E}[Y]$.
- Uncertainty is the background difficulty of the dataset.
- Resolution is helpful separation across forecast groups, so it is subtracted.
- Reliability is calibration error, so it is added.

Main message: a better Brier score can come from better calibration, better separation of easy and hard cases, or both.

Murphy decomposition: what are C , Q , and $\bar{y}$?

Objects

- $Y \in \{0, 1\}$: realized outcome.
- $C \in [0, 1]$: announced probability forecast.
- $Q = \mathbb{E}[Y \mid C] = P(Y = 1 \mid C)$: true event rate inside one forecast group.
- $\bar{y} = \mathbb{E}[Y]$: overall positive rate.

Why Q is the right object

Murphy decomposition groups examples by what the model predicted, not by the original feature vector.

- If many samples receive forecast $C \approx 0.8$, they form one forecast group.
- The relevant truth inside that group is $Q = P(Y = 1 \mid C)$.
- This is why Murphy uses Q , not $\eta = P(Y = 1 \mid X = x)$.

Reading rule: compare the announced forecast C to the actual frequency Q among cases with that same forecast.

Murphy decomposition: first derivation step

Start from the Brier score

$$\text{BS} = \mathbb{E}_{C,Y}[(C - Y)^2] = \mathbb{E}_C[\mathbb{E}_{Y|C}[(C - Y)^2 | C]].$$

Evaluate the conditional expectation

Keep the roles separate: $C \in [0, 1]$ is fixed after conditioning, while $Y \in \{0, 1\}$ is still random.

$$\mathbb{E}[(C - Y)^2 | C] = (C - 1)^2 P(Y = 1 | C) + C^2 P(Y = 0 | C).$$

With $Q = P(Y = 1 | C)$ and $P(Y = 0 | C) = 1 - Q$,

$$\mathbb{E}[(C - Y)^2 | C] = Q(1 - C)^2 + (1 - Q)C^2 = (C - Q)^2 + Q(1 - Q).$$

Expectation bookkeeping: the outer expectation averages over forecast values C , while the conditional expectation averages over the random outcome Y within one forecast group.

$$\Rightarrow \text{BS} = \mathbb{E}_C[(C - Q)^2] + \mathbb{E}_C[Q(1 - Q)].$$

Murphy decomposition: second derivation step

Rewrite the remaining term

$$\begin{aligned}\bar{y} &= \mathbb{E}_Y[Y] \\ &= \mathbb{E}_C[\mathbb{E}_{Y|C}[Y \mid C]] = \mathbb{E}_C[Q], \\ \text{Var}(Q) &= \mathbb{E}[Q^2] - \mathbb{E}[Q]^2 = \mathbb{E}[Q^2] - \bar{y}^2 \quad \Rightarrow \quad \mathbb{E}[Q^2] = \text{Var}(Q) + \bar{y}^2.\end{aligned}$$

$$\begin{aligned}\mathbb{E}[Q(1 - Q)] &= \mathbb{E}[Q] - \mathbb{E}[Q^2] \\ &= \bar{y} - (\text{Var}(Q) + \bar{y}^2) \\ &= \bar{y}(1 - \bar{y}) - \text{Var}(Q).\end{aligned}$$

$$\Rightarrow \quad \text{BS} = \bar{y}(1 - \bar{y}) - \text{Var}(Q) + \mathbb{E}[(Q - C)^2].$$

Bridge to the earlier Brier slide: same binary Brier algebra as before, but now the fixed forecast a becomes the random forecast C , and the truth term becomes $Q = P(Y = 1 \mid C)$ inside each forecast group.

How to interpret uncertainty, resolution, and reliability

Uncertainty

$$\bar{y}(1 - \bar{y})$$

Depends only on the base rate. A 50–50 problem is intrinsically harder than one with mostly 0s or mostly 1s.

Resolution

$$\text{Var}(Q)$$

Large when forecast groups truly have different event rates. This is helpful separation, so it is subtracted.

Reliability

$$\mathbb{E}[(Q - C)^2]$$

Mean squared gap between announced probabilities and actual frequencies within forecast groups. Zero means perfect calibration.

Useful contrast: a constant forecaster can be perfectly reliable but have zero resolution; a sharp forecaster can have high resolution but still be unreliable if its probabilities are numerically wrong.

Expected calibration error (ECE)

Definition

For bins $B_1, \dots, B_M$ and confidence–correctness pairs (c_i, r_i) ,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|,$$

where

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} r_i, \quad \text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} c_i.$$

- **Symbols:** c_i is top confidence; $r_i = 1\{\hat{y}_i = y_i\}$ is correctness.
- **Interpretation:** weighted average of binwise confidence–accuracy gaps.
- **Why it matters:** it is the most common calibration summary in deep-learning practice.

Algorithm: choose bins; place examples; compute mean confidence and mean accuracy per bin; average the weighted absolute gaps.

Worked ECE on the running example (Example B)

Use 3 equal-width bins: $[0, \frac{1}{3})$, $[\frac{1}{3}, \frac{2}{3})$, $[\frac{2}{3}, 1]$.

Nonempty bin	Mean conf.	Mean acc.	Contribution
$[\frac{1}{3}, \frac{2}{3})$	$(0.595 + 0.426 + 0.500)/3 = 0.507$	$0/3 = 0$	$\frac{3}{8} 0 - 0.507 = 0.190$
$[\frac{2}{3}, 1]$	$(0.894 + 0.696 + 0.777 + 0.812 + 0.746)/5 = 0.785$	$4/5 = 0.8$	$\frac{5}{8} 0.8 - 0.785 = 0.009$

$$\text{ECE} \approx 0.190 + 0.009 = 0.199.$$

- Same model, same labels, same confidences.
- Change the bins and the number changes, even though the model did not.

ECE: use it, but do not use it alone

Main caveats

- Depends on the binning rule and the estimator.
- Standard top-label ECE ignores non-top probabilities.
- A small global ECE can hide subgroup failure.
- ECE is not a proper score and can reward bland forecasts.

Bottom line: ECE is a useful diagnostic, not a standalone definition of probability quality.

Safe reporting bundle

- State the calibration notion and ECE variant.
- Pair ECE with NLL or Brier score.
- Show a reliability diagram.
- Keep the train / calibration / test split explicit.

Example C: same accuracy, different confidence quality

Model	Accuracy	Coarse 2-bin ECE	Brier	NLL
A	$5/8 = 0.625$	0.10	0.2075	0.589
B	$5/8 = 0.625$	0.00	0.2213	0.633

- Both models get the same labels right.
- Under coarse bins $[0, 0.5)$ and $[0.5, 1]$, Model B looks perfectly calibrated because all its forecasts collapse into one average bin.
- Proper scores still prefer Model A because it preserves more useful confidence variation.

Lesson: ECE can reward trivial blandness; proper scores and calibration summaries answer different questions.

Generic post-hoc calibration problem

Template

Freeze the base model and learn a map

g : old outputs $\longrightarrow$ new probabilities

by minimizing a holdout-set objective,

$$\arg \min_{g \in \mathcal{G}} \sum_{i=1}^m \ell(g(s_i), y_i).$$

- **Symbols:** s_i can be a scalar score, a logit vector, or a probability vector.
- **Interpretation:** we repair probabilities *after* training, without retraining the classifier.
- **Why it matters:** this is the standard practical response to miscalibrated neural networks.

Design choices: input object, function family, and fitting objective.

Default first baseline: temperature scaling.

Temperature scaling: mechanism and intuition

Definition

$$p_k^{(T)}(x) = \frac{e^{z_k(x)/T}}{\sum_j e^{z_j(x)/T}}, \quad T > 0.$$

- **Symbols:** $z_k(x)$ are logits; T is the temperature.
 - **Interpretation:** $T > 1$ softens the distribution; $T < 1$ sharpens it.
 - **Why it matters:** it changes confidence values with only one parameter.
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- Representative baseline: Guo et al. (2017).
 - Works surprisingly well in many deep networks.

Toy logits example

T	p_A	p_B	p_C
0.5	0.992	0.007	0.001
1.0	0.894	0.073	0.033
2.0	0.676	0.194	0.130

For logits (3.5, 1.0, 0.2):

- larger T pulls mass toward the middle;
- smaller T makes the winner more dominant.

Temperature scaling objective

Fit T by minimizing calibration-set NLL:

$$\mathcal{L}(T) = - \sum_{i=1}^m \log p_{i,y_i}^{(T)} = \sum_{i=1}^m \left(-\frac{z_{i,y_i}}{T} + \log \sum_k e^{z_{i,k}/T} \right).$$

Why the two forms are equivalent

For the temperature-scaled softmax,

$$p_{i,y_i}^{(T)} = \frac{e^{z_{i,y_i}/T}}{\sum_k e^{z_{i,k}/T}},$$
$$-\log p_{i,y_i}^{(T)} = -\log \frac{e^{z_{i,y_i}/T}}{\sum_k e^{z_{i,k}/T}} = -\frac{z_{i,y_i}}{T} + \log \sum_k e^{z_{i,k}/T}.$$

- **Symbols:** (z_i, y_i) are calibration-set logits and labels, and $p_{i,y_i}^{(T)}$ is the post-temperature probability of the true class.
- **Interpretation:** each example contributes “negative true-class score + log normalizer.”

Why optimizing temperature is easy

Reparameterize by $s = 1/T$ and write the one-example loss as

$$\ell_i(s) = -sz_{i,y_i} + \log \sum_k e^{sz_{i,k}}, \quad \mathcal{L}(s) = \sum_{i=1}^m \ell_i(s).$$

Define the temperature-scaled class probabilities

$$p_{i,k}^{(s)} = \frac{e^{sz_{i,k}}}{\sum_j e^{sz_{i,j}}}.$$

Then

$$\ell'_i(s) = -z_{i,y_i} + \frac{\sum_k z_{i,k} e^{sz_{i,k}}}{\sum_j e^{sz_{i,j}}} = -z_{i,y_i} + \sum_k p_{i,k}^{(s)} z_{i,k}.$$

Differentiate once more:

$$\ell''_i(s) = \sum_k p_{i,k}^{(s)} z_{i,k}^2 - \left(\sum_k p_{i,k}^{(s)} z_{i,k} \right)^2 = \text{Var}_{p_i^{(s)}}[z_{i,k}].$$

Therefore

$$\frac{d^2 \mathcal{L}}{ds^2} = \sum_{i=1}^m \ell''_i(s) = \sum_{i=1}^m \text{Var}_{p_i^{(s)}}[z_{i,k}] \geq 0.$$

Bottom line: the objective is convex in $s = 1/T$, so standard one-dimensional optimization finds the global optimum.

What temperature scaling preserves – and what it cannot fix

Argmax preservation

If $z_a(x) \geq z_b(x)$ for every class b , then for any $T > 0$,

$$\frac{z_a(x)}{T} \geq \frac{z_b(x)}{T}.$$

Therefore the same class remains largest after softmax.

- **Interpretation:** temperature scaling changes *numerical honesty*, not class ordering.
- It preserves the predicted label and every within-example class ranking.
- Monotone score rescaling also leaves example ranking unchanged for selective classification.

Limitation

Temperature scaling cannot repair bad **ranking** for abstention or retrieval. It can improve the numbers without moving the risk–coverage curve much.

Post-hoc families and when to use them

Family	Input	Core map	Best use / main caution
Platt scaling	scalar score	one logistic S-curve	simple binary baseline; limited flexibility
Isotonic regression	scalar score	monotone stepwise map	flexible with enough holdout data; can overfit small calibration sets
Beta calibration	scalar probability	logistic model on $\log s$ and $\log(1 - s)$	corrects asymmetric distortions near 0 and 1
Temperature scaling	logit vector	divide all logits by one scalar T	first deep-net baseline; preserves argmax and ranking
Dirichlet calibration	full probability vector	logistic model on log-probability features	use when the whole vector matters; more data hungry

- Start with **temperature scaling** for standard deep classifiers.
- Move to **scalar nonlinear** methods when one score is visibly distorted.
- Move to **vector-aware** methods when downstream code uses more than the top confidence.

Modern extensions of post-hoc calibration

Recent work mainly changes post-hoc calibration for one of four deployment questions:

- **Many classes, top confidence only:** calibrate the event “the top prediction is correct” directly when full-vector calibration is unnecessary.
- **Ranking-sensitive use:** regularize flexible scalar calibrators so probability repair does not badly damage discrimination.
- **Full-vector use:** use multiclass maps that repair the whole probability simplex, often while preserving the top label.
- **Shifted deployment:** combine recalibration with label-shift or target-shift estimates.

Representative directions: many-class confidence calibration, ROC-aware isotonic calibration, accuracy-preserving multiclass calibration, and LaSCal-style shift-aware recalibration.

Theme: modern variants are task-matched extensions of the same post-hoc idea, not a new paradigm.

Do not conflate three different interventions

Intervention	What it changes	Typical question
Post-hoc calibration	probability values after training	Are the numbers numerically honest?
Uncertainty estimation (ensembles, MC dropout)	the predictive aggregate itself	Do we have a stronger uncertainty signal than one softmax pass?
Selective prediction	the final action rule	Given a score, when should the system answer or abstain?

Bottom line: calibration repair, uncertainty estimation, and abstention solve related but different problems.

Deep ensembles: mechanism and why averaging helps

$$\bar{p}(x) = \frac{1}{M} \sum_{m=1}^M p^{(m)}(x).$$

- **Symbols:** $p^{(m)}(x)$ is the probability vector from independently trained model m .
- **Interpretation:** the ensemble averages several plausible predictors instead of trusting one training run.
- **Why it matters:** this often improves uncertainty quality, especially on ambiguous or shifted inputs.

For a convex score $S(\cdot, y)$,

$$S(\bar{p}, y) \leq \frac{1}{M} \sum_{m=1}^M S(p^{(m)}, y).$$

- Jensen's inequality explains why averaging can smooth overconfident spikes.
- **Strength:** strong empirical uncertainty baseline.
- **Limitation:** requires training and storing multiple full models.

MC dropout

Operational algorithm

Keep dropout active at test time, run T stochastic forward passes, and average:

$$\bar{p}(x) = \frac{1}{T} \sum_{t=1}^T p^{(t)}(x).$$

- **Symbols:** $p^{(t)}(x)$ uses a different dropout mask on pass t .
 - **Interpretation:** approximate an ensemble by sampling many subnetworks.
 - **Why it matters:** cheaper retrofit than training many independent models.
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- Bayesian view: Gal and Ghahramani (2016) interpret dropout masks as a Bernoulli variational family.
 - Main strength: easy to add to an existing network with dropout.
 - Main limitation: usually weaker than deep ensembles as an uncertainty baseline.

Predictive entropy, variance, and mutual information

Given stochastic predictions $p^{(1)}(x), \dots, p^{(T)}(x)$,

$$\bar{p}(x) = \frac{1}{T} \sum_{t=1}^T p^{(t)}(x), \quad H(\bar{p}) = - \sum_k \bar{p}_k \log \bar{p}_k.$$

$$\widehat{\text{Cov}}(x) = \frac{1}{T} \sum_{t=1}^T (p^{(t)} - \bar{p})(p^{(t)} - \bar{p})^\top,$$

$$\text{MI}(x) = H(\bar{p}) - \frac{1}{T} \sum_{t=1}^T H(p^{(t)}).$$

- **Interpretation:** entropy measures total uncertainty; covariance and mutual information measure disagreement across sampled predictors.
- **Why it matters:** disagreement is often a useful proxy for epistemic uncertainty.

Practical comparison: what should you try first?

Method	Extra cost	training	Test-time cost	Main use / main limitation
Temperature scaling	tiny		tiny	best first calibration baseline; improves numbers but not ranking
Deep ensembles	high		moderate–high	strong uncertainty baseline, often strong under shift and abstention; expensive
MC dropout	low–moderate		moderate	cheaper predictive averaging; usually weaker than full ensembles

- Ding, Cao, and Luo (2023): ensembles help selective classification most on *ambiguous* points near decision boundaries.
- Good default workflow: temperature scaling first, then compare against deep ensembles if abstention or shift matters.

Reject option from the simplest cost model

Suppose we can either predict a class or abstain.

- Wrong prediction costs 1.
- Abstention costs $\lambda \in (0, 1)$.

If we answer with the most likely class, the conditional expected loss is

$$1 - \max_k P(Y = k \mid X = x).$$

Why this is the right expression

Let

$$a^*(x) = \arg \max_k P(Y = k \mid X = x).$$

Under 0–1 loss, answering with $a^*(x)$ has conditional expected loss

$$\mathbb{E}[1\{Y \neq a^*(x)\} \mid X = x] = P(Y \neq a^*(x) \mid X = x) = 1 - P(Y = a^*(x) \mid X = x) = 1 - \max_k P(Y = k \mid X = x).$$

If we abstain, the loss is λ . Therefore answer exactly when

$$1 - \max_k P(Y = k \mid X = x) \leq \lambda \iff \max_k P(Y = k \mid X = x) \geq 1 - \lambda.$$

Decision rule: answer only when the best achievable error at x is no larger than the reject cost.

Reject threshold: intuition and example

Suppose

$$P(Y = A \mid X = x) = 0.7, \quad P(Y = B \mid X = x) = 0.2, \quad P(Y = C \mid X = x) = 0.1.$$

Then the Bayes action is to answer class A, and the conditional expected loss of answering is

$$1 - \max_k P(Y = k \mid X = x) = 1 - 0.7 = 0.3.$$

$$\lambda < 0.3 \Rightarrow \text{abstain}, \quad \lambda \geq 0.3 \Rightarrow \text{answer}.$$

- **Interpretation:** answer only when the expected error of answering is no worse than the reject cost.
- **Operational view:** this becomes a confidence threshold $\max_k P(Y = k \mid X = x) \geq 1 - \lambda$.
- **Why it matters:** this is the basic decision-theoretic justification for thresholding confidence.

Coverage and selective risk

Definitions for a score threshold τ

If we answer on the set $\{x : s(x) \geq \tau\}$, then

$$\text{Coverage}(\tau) = \mathbb{P}(s(X) \geq \tau), \quad R_{\text{sel}}(\tau) = \mathbb{P}(\hat{Y} \neq Y \mid s(X) \geq \tau).$$

- **Symbols:** $s(x)$ is a confidence or uncertainty score.
 - **Interpretation:** coverage is how often we answer; selective risk is the error rate on the answered cases.
 - **Why it matters:** selective systems are judged by both how many cases they keep and how safe those kept cases are.
-
- Ordinary risk is the special case at coverage 1.
 - In deployment, acceptable operating points are usually coverage-constrained.

Risk–coverage curve on the running example (Example B)

Sort Example B by decreasing confidence:

$(0.894, 1), (0.812, 1), (0.777, 1), (0.746, 0), (0.696, 1), (0.595, 0), (0.500, 0), (0.426, 0),$

where each pair is (confidence, correctness).

Accepted top k	Coverage $k/8$	Errors among accepted	Selective risk
8	1.000	4	0.500
6	0.750	2	0.333
5	0.625	1	0.200
3	0.375	0	0.000

- For each row, accept only the first k pairs in the sorted list.
- Coverage = $k/8$, errors = $\sum_{i=1}^k (1 - r_i)$, and selective risk = $\frac{\text{errors among accepted}}{k}$.
- Example: for $k = 5$, the accepted correctnesses are $(1, 1, 1, 0, 1)$, so errors = 1 and selective risk = $1/5 = 0.200$.

Calibration and ranking are related but different

Monotone recalibration preserves ranking

If g is monotone increasing, then

$$s_i \geq s_j \iff g(s_i) \geq g(s_j).$$

- **Interpretation:** a monotone calibrator can fix numerical honesty without changing which examples are accepted first.
- **Why it matters:** selective prediction depends crucially on **ranking quality**, not only calibration.
- Ranking-aware evaluation is emphasized by recent work: Pugnana–Ruggieri (AUC-style summaries), Traub et al. (AUGRC), and Rabanser–Papernot (selective-classification gap).

Practical lesson

If the risk–coverage curve is the deployment objective, test the score ordering directly.
Temperature scaling alone may barely move that curve.

Method families for abstention

Family	Core mechanism	What is optimized or adapted	Strength / main limitation
Post-hoc thresholding	choose a score $s(x)$ and abstain below a threshold τ	only the final decision rule	easiest deployment path; limited by whatever ranking the score already has
SelectiveNet	learn a prediction head f_θ and a selection head g_θ jointly, usually with a coverage penalty	representation, predictor, and reject rule together	can improve abstention ranking directly; needs end-to-end retraining and a target coverage level
Deep Gamblers	add a reservation output and let the network keep probability mass for abstention	the output simplex itself includes a reject option	elegant end-to-end abstention mechanism; behavior depends on the utility or loss design
OOD-aware / plugin methods	estimate both correctness and whether the point is in-distribution	the abstention score, often together with an OOD detector	useful in open-world deployment; more moving parts than closed-world thresholding

Organizing principle: thresholding assumes the score is already good; end-to-end methods try to *learn a better ordering of safe versus risky examples*.

How selective systems should be reported

- Report the **score** used for abstention.
- Show the full risk–coverage curve, not only one operating point.
- Report selective risk at a few target coverages such as 95%, 90%, and 80%.
- If users see confidence values, also report calibration on the **accepted set**.
- Stress-test the curve under plausible distribution shift or OOD contamination.
- If a scalar summary is required, say which one and why: AURC, AUGRC, AUC-style ranking metric, etc.

Bottom line: a selective classifier is a **decision system**; evaluation must reflect the operating trade-off, not only raw calibration.

Why deployment breaks calibration

Shift type	What changes?	Typical reliability failure
Covariate shift	$P(X)$ changes; label semantics stay the same	confidences no longer match correctness on the new input mix
Label shift	class priors $P(Y)$ change	source-calibrated probabilities become numerically wrong in the target population
OOD inputs	deployment includes unseen kinds of inputs	model can be confidently wrong because it is forced to choose among known classes

- Calibration is a conditional statement about the joint distribution of predictions and outcomes.
- If that joint distribution changes, calibration can change even when the model weights stay fixed.
- Ovadia et al. (2019) and Minderer et al. (2021) made this fragility vivid for modern deep models.

Label-shift correction formula

Identity under label shift

Assume $P_t(X | Y) = P_s(X | Y)$ but class priors differ between source s and target t . Then

$$P_t(Y = y | X = x) = \frac{w_y P_s(Y = y | X = x)}{\sum_j w_j P_s(Y = j | X = x)}, \quad w_y = \frac{P_t(Y = y)}{P_s(Y = y)}.$$

- **Symbols:** w_y reweights source posteriors by the change in class priors.
- **Interpretation:** even a source-calibrated classifier can become miscalibrated when the label mix changes.
- **Why it matters:** calibration is distribution-specific, not a once-and-for-all property.

Practical lesson: if deployment priors differ, source-domain temperature scaling may not be enough.

Global averages can hide subgroup failure

- A small global ECE does *not* imply every class or subgroup is reliable.
- Modern stronger notions ask calibration to hold on more granular collections of events.

Stronger notion	What it asks for
Local calibration	reliability should hold in local neighborhoods of score or feature space
Multicalibration	reliability should hold simultaneously across many identifiable subgroups
Strong calibration testing	audit whether <i>some</i> subgroup still has large residual miscalibration

Representative works: Hébert-Johnson et al. (2018), Luo et al. (2022), Feng et al. (2024).

A small global number can hide a bad subgroup

Suppose two equally large subgroups are evaluated separately.

Group	Mean conf.	Empirical acc.	Gap
A	0.90	0.70	+0.20
B	0.50	0.70	-0.20

Globally, the mean confidence is 0.70 and the mean accuracy is also 0.70.

- A single global calibration number can therefore look perfect.
- Yet Group A is substantially overconfident and Group B is substantially underconfident.
- This is exactly the failure mode that motivates local calibration, multicalibration, and strong subgroup testing.

Bottom line: subgroup reliability is not a cosmetic refinement; it prevents important failures from averaging away.

Recent literature: four questions the field now asks

Question	Main move	Why it matters / representatives
Are our calibration metrics trustworthy?	replace crude binning by distribution matching, stronger testing, or truthfulness analyses	calibration auditing itself can be biased or gameable; Marx et al., Hu et al., Feng et al., Haghtalab et al.
What notion of calibration matches the task?	separate top-confidence, full-vector, many-class, and subgroup reliability	the right target depends on what the downstream user sees or uses; Le Coz et al., Esaki et al., Chidambaram–Ge
How should calibration change under shift?	insert label-shift or target-shift estimates into the recalibration step	source-domain reliability is not deployment-domain reliability; Popordanoska et al., Li–Sur
What makes a selective classifier good?	evaluate ranking directly and design losses that improve safe-versus-risky ordering	abstention is about ordering, not only numeric honesty; Pugnana–Ruggieri, Traub et al., Mao et al., Rabanser–Papernot

Theme: recent work keeps the classical pipeline but makes each stage more realistic: better metrics, task-matched calibration targets, shift-aware repair, and ranking-aware abstention.

A practical reliability workflow

- 1 Train a probabilistic classifier with cross-entropy or another proper-score-based objective.
- 2 Report accuracy *and* a proper score such as NLL or Brier.
- 3 Audit calibration with a reliability diagram and a clearly specified ECE variant.
- 4 Fit temperature scaling on a clean calibration split as the first post-hoc baseline.
- 5 If uncertainty quality matters, compare against deep ensembles or MC dropout.
- 6 If the system can abstain, evaluate the score ranking with risk–coverage curves and operating points.
- 7 Stress-test under plausible shift and inspect subgroup reliability, not only global averages.

Final takeaway

Reliable classification is best understood as

probability estimation $\rightarrow$ scoring $\rightarrow$ repair $\rightarrow$ uncertainty $\rightarrow$ decision.

A trustworthy system needs the whole chain, not just high top-1 accuracy.

Appendix: the binary weather table from Example A

Day	Forecast p	Outcome y	$(p - y)^2$	Log-loss term
1	0.90	1	0.01	$-\log 0.90 \approx 0.105$
2	0.70	0	0.49	$-\log 0.30 \approx 1.204$
3	0.40	1	0.36	$-\log 0.40 \approx 0.916$
4	0.20	0	0.04	$-\log 0.80 \approx 0.223$
Average	–	–	BS = 0.225	NLL ≈ 0.612

- This is the smallest worked example in the source.
- It makes the role of Brier score and log loss concrete before moving to multiclass logits and softmax.

Appendix: full multiclass toy example (Example B)

#	Logits $z(x)$ and softmax $p(x)$	Pred.	Conf.	True	Correct?
1	$z = (3.5, 1.0, 0.2), p = (0.894, 0.073, 0.033)$	A	0.894	A	1
2	$z = (2.9, 1.7, 0.9), p = (0.696, 0.210, 0.094)$	A	0.696	A	1
3	$z = (1.0, 2.8, 0.7), p = (0.128, 0.777, 0.095)$	B	0.777	B	1
4	$z = (3.0, 2.5, 0.4), p = (0.595, 0.361, 0.044)$	A	0.595	B	0
5	$z = (0.2, 0.8, 2.7), p = (0.067, 0.121, 0.812)$	C	0.812	C	1
6	$z = (3.2, 1.9, 0.5), p = (0.746, 0.203, 0.050)$	A	0.746	B	0
7	$z = (1.8, 1.5, 1.3), p = (0.426, 0.316, 0.258)$	A	0.426	C	0
8	$z = (0.5, 2.2, 2.0), p = (0.091, 0.500, 0.409)$	B	0.500	C	0

- Accuracy = 0.50, mean top confidence ≈ 0.681 .
- The source also reports multiclass BS ≈ 0.461 and NLL ≈ 0.724 on this toy set.

Appendix: detailed ECE calculation (Example B)

Using equal-width bins $[0, \frac{1}{3})$, $[\frac{1}{3}, \frac{2}{3})$, $[\frac{2}{3}, 1]$:

Bin	Confidences	Correctness	Size	Mean conf.	Mean acc.
$[\frac{1}{3}, \frac{2}{3})$	0.595, 0.426, 0.500	0, 0, 0	3	0.507	0.000
$[\frac{2}{3}, 1]$	0.894, 0.696, 0.777, 0.812, 0.746	1, 1, 1, 1, 0	5	0.785	0.800

$$\text{ECE} = \frac{3}{8}|0.507 - 0| + \frac{5}{8}|0.785 - 0.800| \approx 0.199.$$

Why this appendix exists: on a first pass, the important lesson is the algorithm; later, the full arithmetic is useful for self-study.

Appendix: why the post-hoc family branches

- **Platt scaling:** $g(s) = \text{sigmoid}(As + B)$ on one scalar score s . This is logistic regression on a single calibration feature.
- **Isotonic regression:** fit a nondecreasing map on sorted scores. The pool-adjacent-violators (PAV) idea is simple: whenever two neighboring blocks violate monotonicity, merge them and replace them by the block-average label.
- **Beta calibration:** $g(s) = \text{sigmoid}(a \log s + b \log(1 - s) + c)$. The $\log s$ and $\log(1 - s)$ features let the map correct asymmetric distortions near 0 and 1.
- **Dirichlet calibration:** $g_k(p) \propto \exp(a_k^\top \log p + b_k)$ with $\log p = (\log p_1, \dots, \log p_K)$. This is the natural vector analogue of beta-style log-probability features.

Relation to temperature scaling

Temperature scaling is the most rigid vector calibrator: it rescales all logits by one scalar. Dirichlet calibration is more flexible because it can couple classes and alter non-top probabilities when the full vector matters.

Appendix: MC dropout and the ELBO chain

Bayesian predictive ideal

$$p(y \mid x, \mathcal{D}) = \int p(y \mid x, W) p(W \mid \mathcal{D}) dW.$$

Variational approximation

Choose a simpler family $q_{\theta}(W)$ and maximize

$$\text{ELBO}(\theta) = \mathbb{E}_{q_{\theta}(W)}[\log p(\mathcal{D} \mid W)] - \text{KL}(q_{\theta}(W) \parallel p(W)).$$

- Dropout masks define a Bernoulli variational family.
- Training with dropout plus weight decay approximates ELBO optimization.
- Test-time stochastic averaging approximates the posterior predictive distribution.

Appendix: four historical branches to remember

Branch	Canonical lineage	Lasting question
Forecast evaluation	Brier → Murphy → Dawid / DeGroot–Fienberg → Gneiting–Raftery	How should probability forecasts be scored and decomposed?
Post-hoc calibration	Platt → Zadrozny–Elkan → beta / temperature / Dirichlet	How do we repair probabilities after training?
Uncertainty approximation	dropout as approximate Bayesian / variational inference → deep ensembles → shift benchmarks	How do we represent model uncertainty better than one softmax pass?
Reject option	Chow → selective classification → SelectiveNet / Deep Gamblers	When should the system abstain rather than answer?

Why this appendix matters: recent papers are much easier to place once you know which historical question they extend.

Appendix: stronger than global calibration

- **Global calibration:** one average relation between confidence and accuracy over the whole population.
- **Local calibration:** the relation should hold in local neighborhoods of score space or feature space, not only after global averaging.
- **Multicalibration:** the relation should hold simultaneously over many identifiable subgroups.
- **Strong calibration testing:** audit whether *some* subgroup still has a large residual gap, even when the global metric looks small.

Why this matters

A system can have a small overall ECE while failing badly on a rare class, a demographic slice, or a clinically important subgroup. Stronger notions ask reliability to survive that finer conditioning.

Appendix: selected references by cluster

Classical foundations

- Brier (1950), Murphy (1973)
- Dawid (1982), DeGroot–Fienberg (1983)
- Gneiting–Raftery (2007)
- Platt (1999), Zadrozny–Elkan (2001, 2002)
- Guo et al. (2017), Kull et al. (2017, 2019)

Uncertainty and shift

- Gal–Ghahramani (2016)
- Lakshminarayanan et al. (2017)
- Ovadia et al. (2019)
- Minderer et al. (2021)

Selective prediction

- Chow (1957, 1970)
- Geifman–El-Yaniv (2017, 2019)
- Liu et al. (2019)
- Mao et al. (2024), Narasimhan et al. (2024)

Recent auditing and evaluation

- Marx et al. (2023), Pugnana–Ruggieri (2023)
- Hu et al. (2024), Feng et al. (2024)
- Haghtalab et al. (2024), Traub et al. (2024)
- Chidambaram–Ge (2025), Li–Sur (2025), Rabanser–Papernot (2025)